7th September 2026



Topological Methods in Condensed-Matter Physics

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Course details

- **Time and location:**
  - lecture/exercise group: Monday 16:15-18:45 F3M01
- **Website:** https://edu.ttk.bme.hu/course/view.php?id=4377
- **Prerequisites:**
  - multivariable calculus
  - linear algebra
  - quantum mechanics
  - introduction to solid-state physics
- **Requirements:**
  - **oral presentation** of a paper (15 minutes) during the last semester week (7th, 14th December)
  - discuss the topological concept from the mathematical point of view and what kind of physical phenomenon it describes
  - consultation with the lecturers recommended before the presentation
  - **oral exam** during the exam period: discussion of theoretical exercises based on the lecture
  - exercises made available on moodle during the semester in 4 problem sheets
  - solutions may be handed in over moodle, feedback received from the lecturers
  - grade based on presentation and oral exam

Presentation topics

	Topic	Paper	Speaker
1	**Magnetic skyrmions**	N. Nagaosa *et al.*, Nat. Nanotechnol. **8**, 899	
2	**Magnetic hopfions**	P. J. Ackerman *et al.*, Nat. Mater. **16**, 426	
3	**Skyrmions with various topological charges**	F. N. Rybakov *et al.*, Phys. Rev. B **99**, 064437	
4	**Topological defects in liquid crystals**	S. Fumeron *et al.*, Eur. Phys. J. Spec. Top. **232**, 1813	
5	**Ferroelectric textures**	Y. J. Wang et al., Acta Mater. **243**, 118485	
6	**Optical skyrmions**	Z. Ye *et al.*, Proc. R. Soc. A, **480**, 20240109	
7	**Topology of two-qubit states**	R. Mosseri *et al.*, J. Phys. A: Math. Gen. **34**, 10243	
8	**Geometry of the Hermitian matrix space**	G. Pintér *et al.*, Quantum **10**, 2047	
9	**Geometric phase**	E. Cohen *et al.*, Nat. Rev. Phys. **1**, 437	

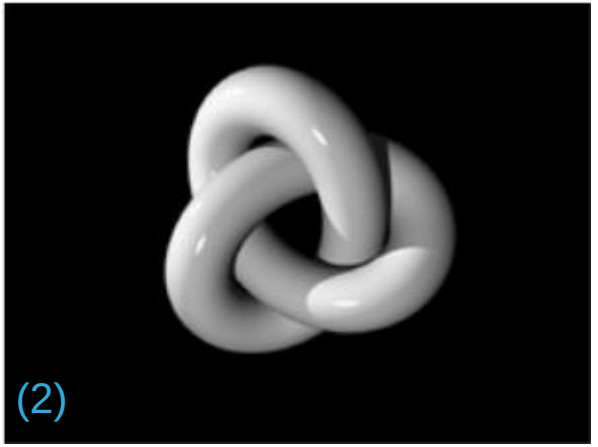
Presentation topics

	Topic	Paper	Speaker
10	**Quantum Hall effect**	D. Tong, The Quantum Hall Effect, Chapter 2	
11	**Topological insulators in 2D**	S. Blügel *et al.*, Topological Matter, Chapter B1	
12	**Topological semimetals**	S. Blügel *et al.*, Topological Matter, Chapter B2	
13	**Topology of time-reversal-invariant systems**	J. E. Moore *et al.*, Phys. Rev. B **75**, 121306	
14	**Hopf insulators**	J. E. Moore *et al.*, Phys. Rev. Lett. **101**, 186805	
15	**Periodic table of topological insulators**	A. Kitaev, AIP Conf. Proc. **1134**, 22	
16	**Topological photonics**	L. Lu *et al.*, Nat. Photon. **8**, 821	
17	**Topological magnons**	P. A. McClarty, Annu. Rev. Condens. Matt. Phys. **13**, 171	
18	**Topological phononics**	Y. Liu *et al.*, Adv. Funct. Mater. **2019**, 1904784	
19	**Topology of non-Hermitian systems**	C. C. Wojcik *et al.*, Phys. Rev. B **101**, 205417	

Topology



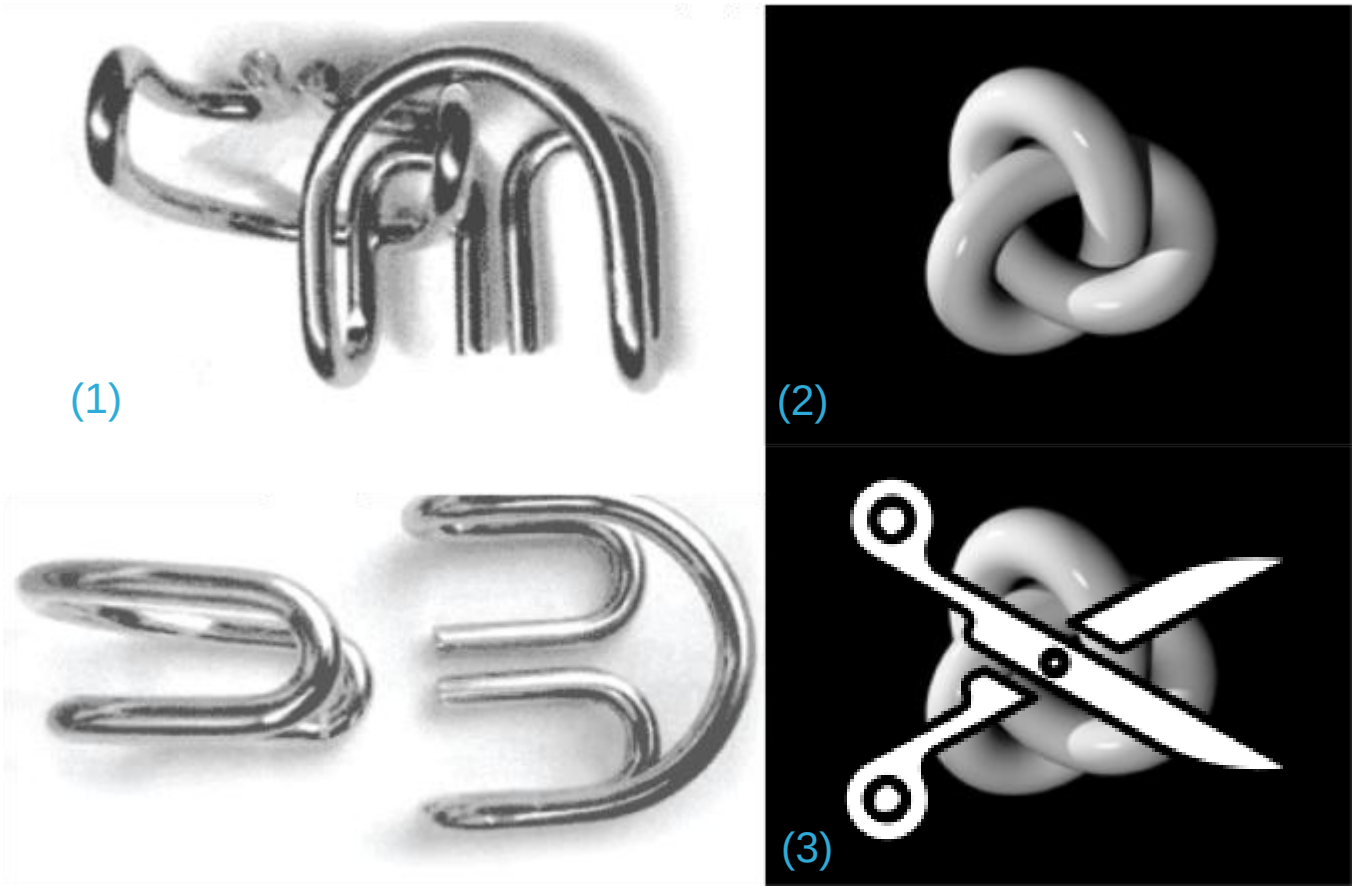
(1)



(2)

- topology describes global properties of objects which do not change under deformations

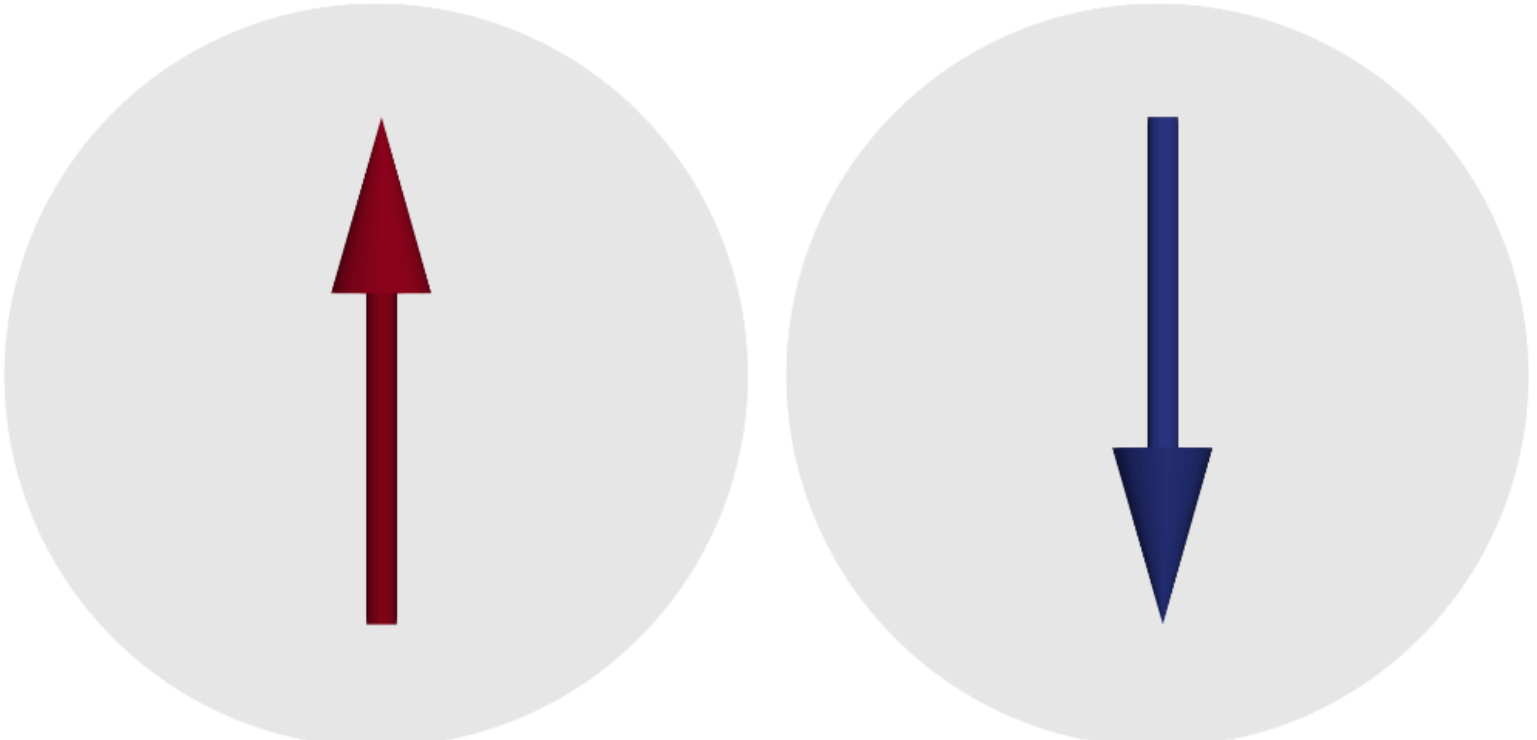
Topology



topologically trivial topologically non-trivial

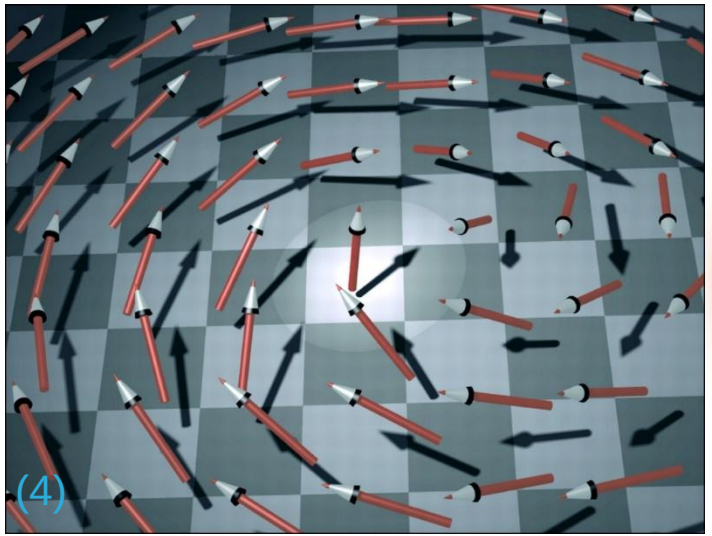
- topology describes global properties of objects which do not change under deformations

Topology in physics

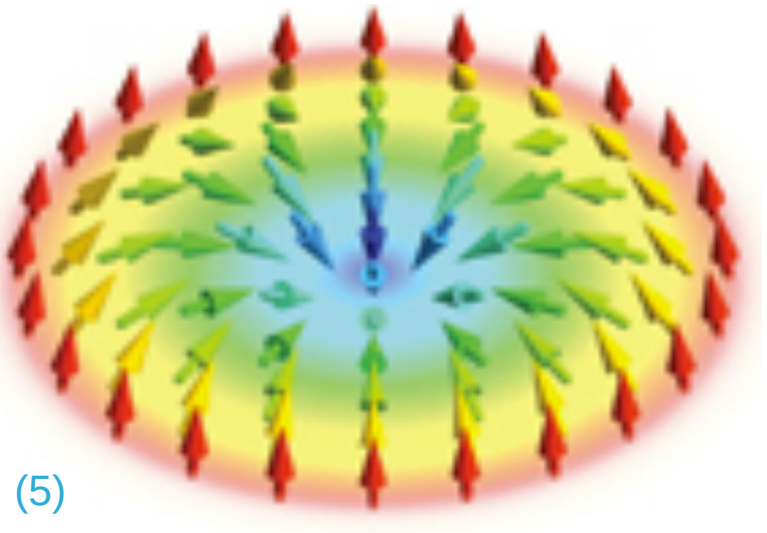


- vector fields with non-vanishing length connected to the topology of spheres
- magnetism, liquid crystals, superconductors, electromagnetic fields, particles, cosmology

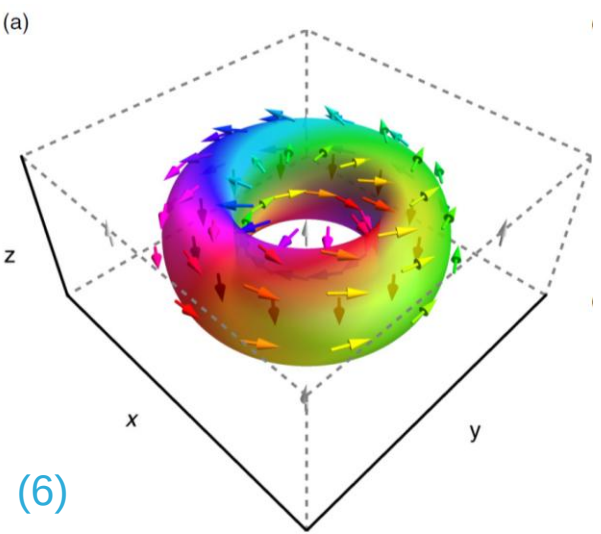
Topology in real space



vortex



skyrmion



hopfion

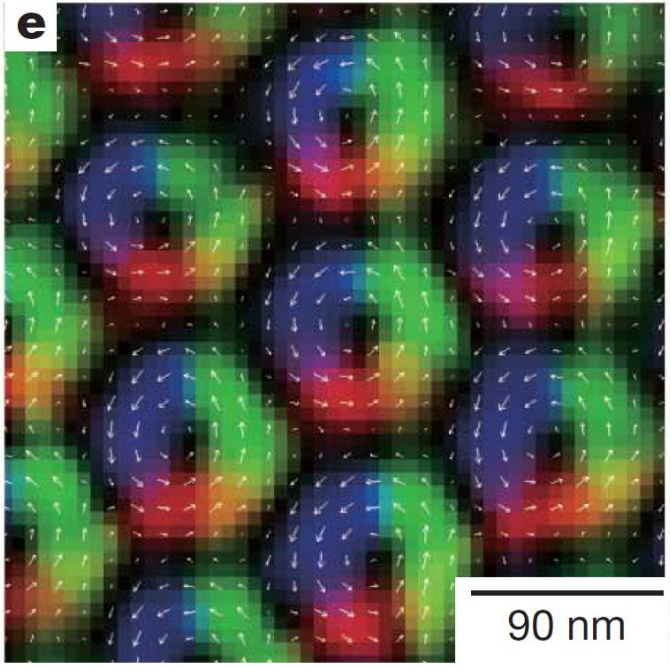
- fields form topological defects robust against deformation

Topology in real space



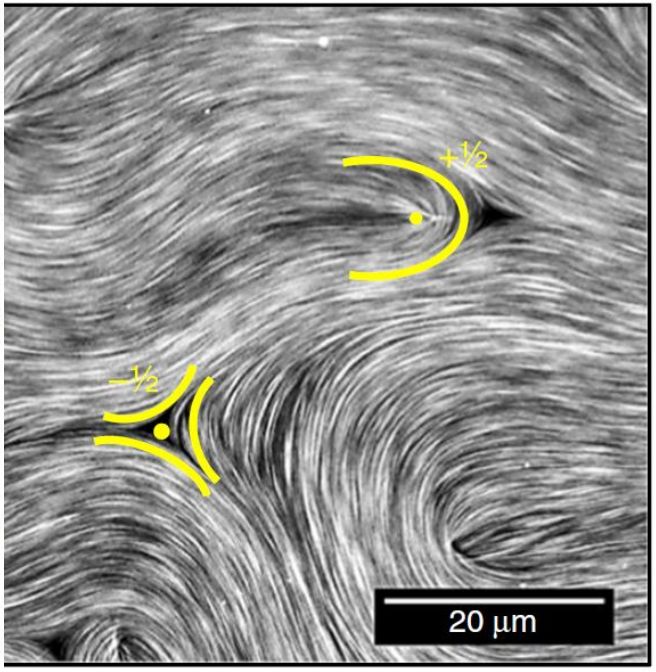
(7)

x-ray circular dichroism



(8)

Lorentz TEM

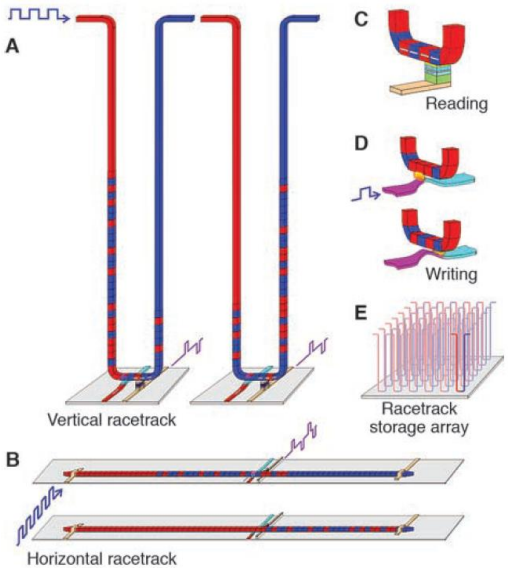


(9)

fluorescence confocal microscopy

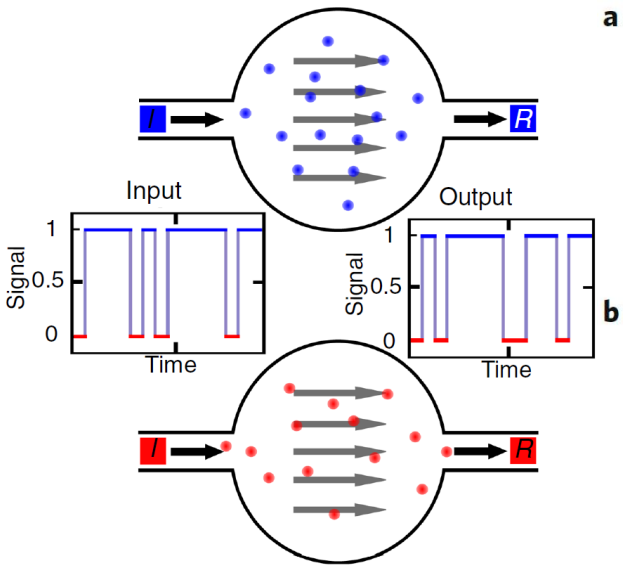
- topological defects may be visualized using imaging techniques

Topology in real space



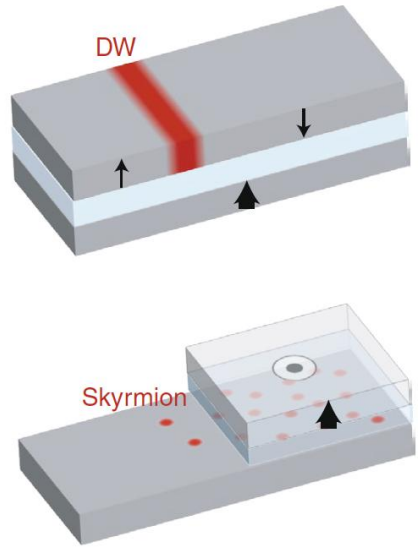
(10)

binary



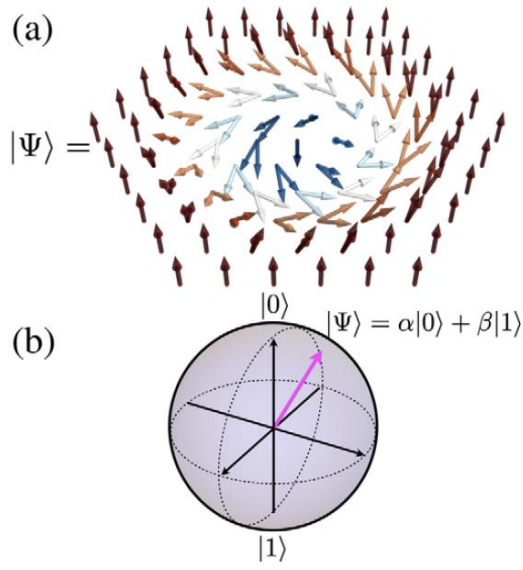
(11)

stochastic



(12)

neuromorphic

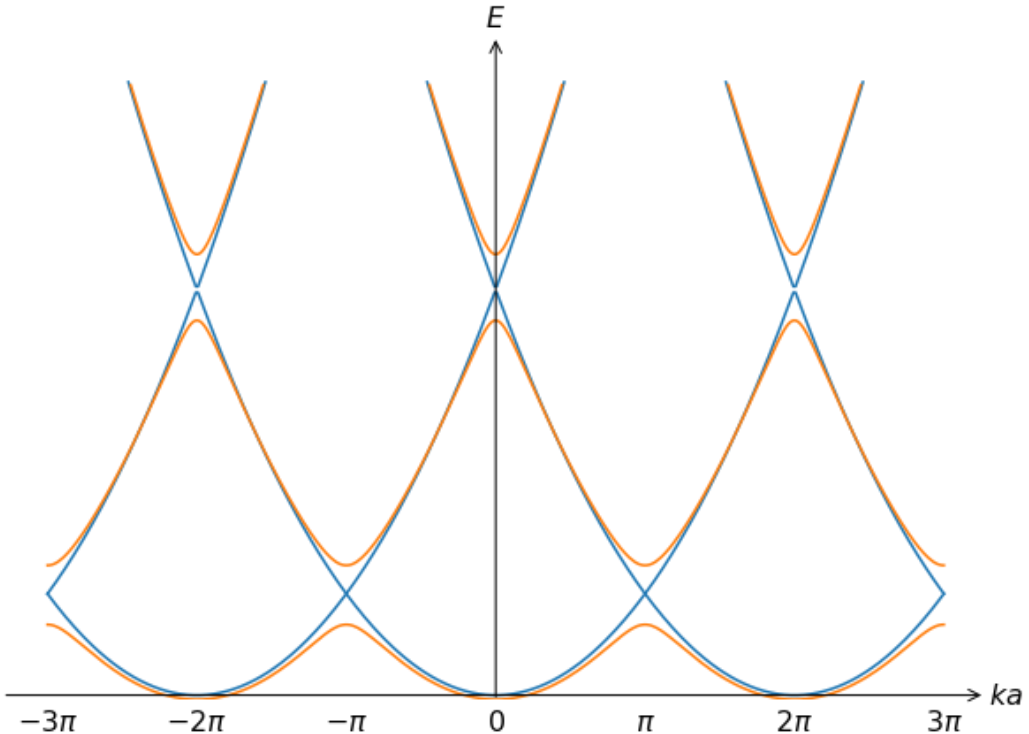


(13)

quantum

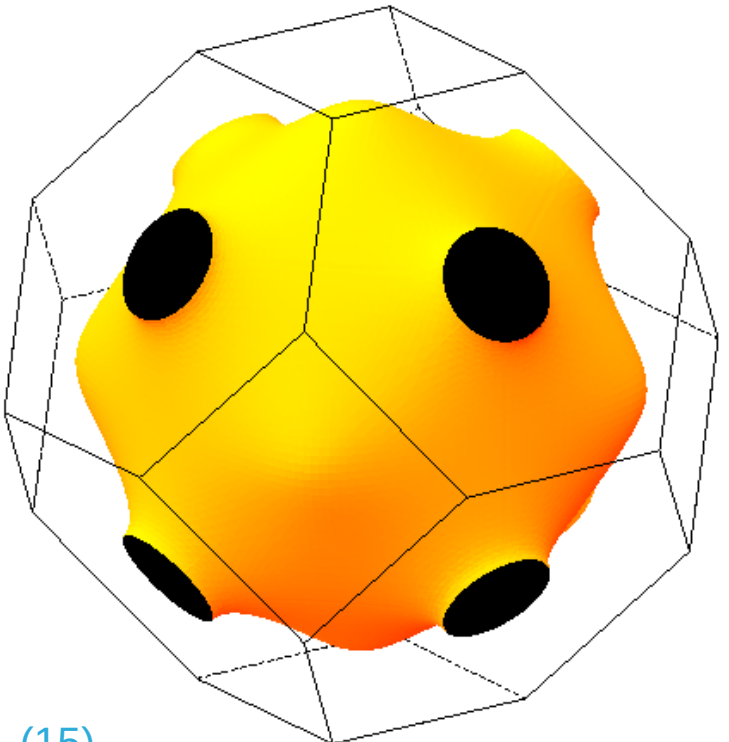
- topological defects inspired various types of applications

Topology in reciprocal space



(14)

dispersion relation

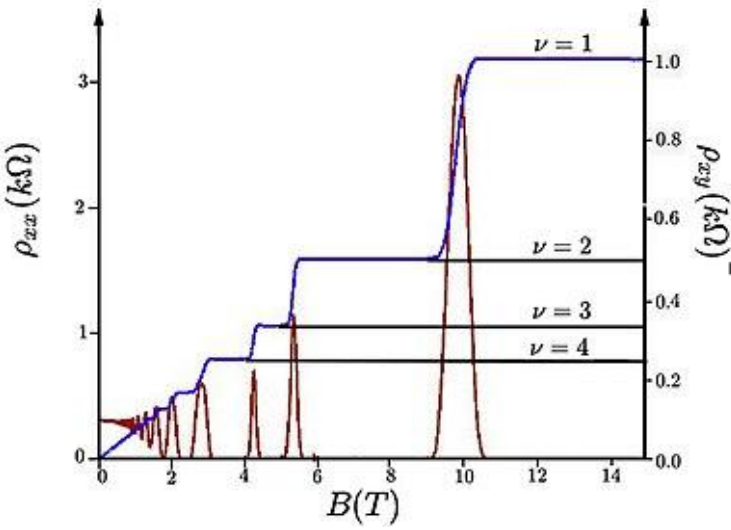


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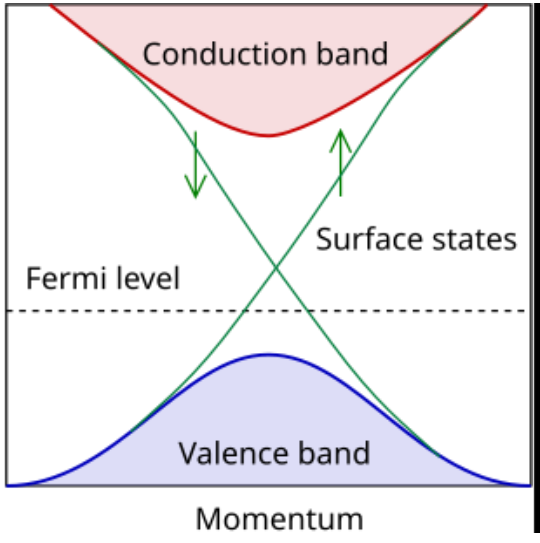
Fermi surface

- crystals are periodic arrangements of atoms, eigenstates may be described in Fourier space
- periodic repetition in Fourier space is topologically described by tori

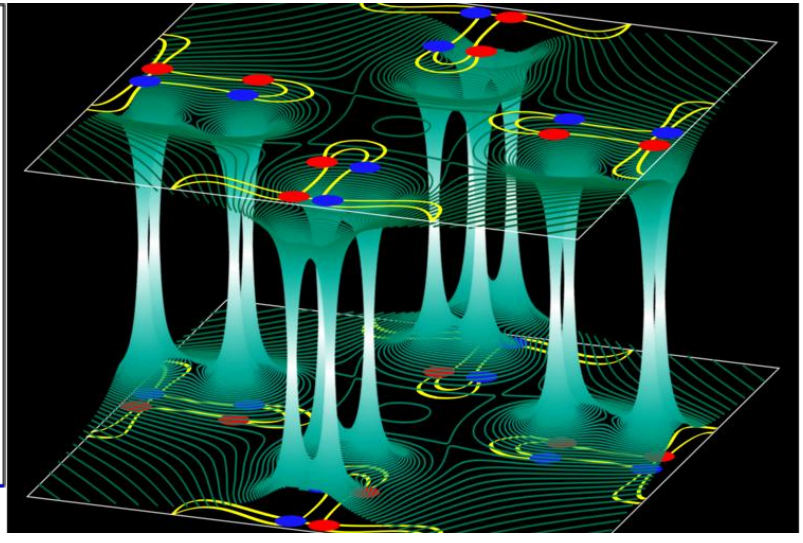
Topology in reciprocal space



(16) quantum Hall effect



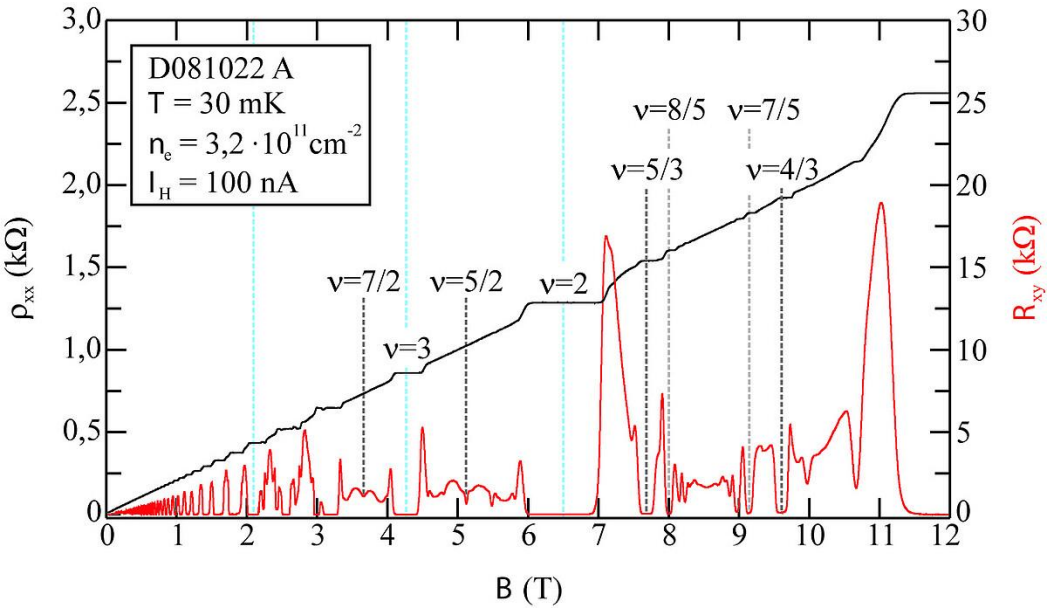
(17) topological insulators



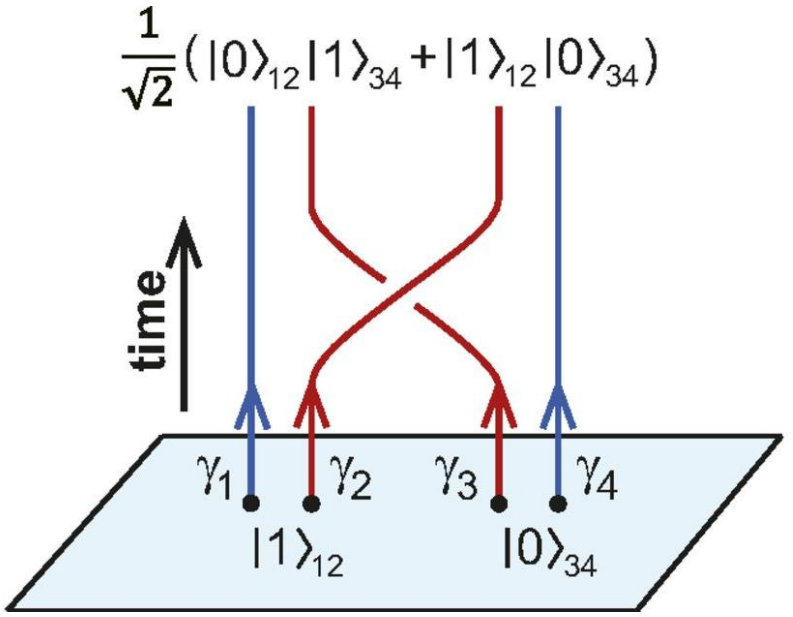
(18) Weyl semimetals

- topological properties may be observed in transport or in the band structure

Exchange statistics and topology



(19) fractional quantum Hall effect



(20) Majorana bound states

- the transformation of eigenstates under exchanging two indistinguishable particles determines physical properties of particles (bosons, fermions)
- the transformation rules depend on the topology of the configuration space (anyons)

Course contents

- introduction to basic mathematical concepts of topology
- application of these concepts to the description of physical phenomena
- **topics:**

I. Topological spaces
   - Construction and examples
   - Homotopy
   - Fixed-length vector fields, Brillouin zone, quantum-mechanical states

II. Manifolds
   - Degree of a mapping
   - Hopf theorem, Poincaré-Hopf theorem
   - Topological defects, topological insulators

III. Vector bundles
   - Euler number, Chern number
   - Connection and curvature
   - Berry phase, connection to topological invariants

Literature

(1) lecture notes
(2) J. Zang, V. Cros, A. Hoffmann, Topology in Magnetism
(3) S. Blügel, Y. Mokrousov, T. Schäpers, Y. Ando, Topological Matter – Topological Insulators, Skyrmions and Majoranas

References

(1) https://www.archimedes-lab.org/How_to_Solve/MW_puzzle.html
(2) https://en.wikipedia.org/wiki/File:Trefoil_knot_arb.png
(3) https://icons-for-free.com/cut+cutter+scissor+scissors+shears+trim+icon-1320185951970648591/
(4) https://www.science.org/doi/10.1126/science.1075302
(5) http://www.christophschuette.com/physics/skyrmions.php
(6) https://journals.aps.org/prresearch/abstract/10.1103/PhysRevResearch.2.013315
(7) https://www.pnas.org/doi/10.1073/pnas.1711652115
(8) https://www.nature.com/articles/nature09124
(9) https://www.nature.com/articles/s41467-018-05666-8
(10) https://www.science.org/doi/10.1126/science.1145799
(11) https://journals.aps.org/prapplied/abstract/10.1103/PhysRevApplied.9.064018
(12) https://www.nature.com/articles/s41928-019-0360-9
(13) https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.127.067201
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(15) https://www.hzdr.de/projects/fermisur/cu.html
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(17) https://en.wikipedia.org/wiki/File:Topological_insulator_band_structure.svg
(18) https://yazdanilab.princeton.edu/projects/weyl-and-dirac-semi-metals
(19) https://www.fkp.uni-hannover.de/en/research-groups/group-haug/research/quantum-hall-effect
(20) https://www.pnas.org/doi/full/10.1073/pnas.1810003115